

$^3\text{He}(^{32}\text{S}, n\gamma)$ 1985A118

1985A118: $^3\text{He}(^{32}\text{S}, n\gamma)$ with 80-MeV ^{32}S beam on 35-keV ^3He -impinged Au foils. Neutrons were detected using an NE213 scintillator and γ rays were detected using 3 Ge(Li) detectors. Measured $n\gamma$ coin. Deduced lifetimes for 4 levels using the Doppler-Shift Attenuation Method (DSAM).

 ^{34}Ar Levels

<u>E(level)</u>	<u>$T_{1/2}$</u>	<u>Comments</u>
0		
2091	319 fs 42	$T_{1/2}$: lifetime $\tau=460$ fs 60; 490 fs 65 (lineshape) and 410 fs 80 (centroid-shift).
3288	194 fs 35	$T_{1/2}$: lifetime $\tau=280$ fs 50; 255 fs 50 (lineshape) and 385 fs 95 (centroid-shift).
3871	>187 fs	$T_{1/2}$: lifetime $\tau>270$ fs (lineshape). >200 fs (centroid-shift).
4128	9×10^1 fs +12-9	$T_{1/2}$: lifetime $\tau=130$ fs +170-130 (centroid-shift). <310 fs (lineshape).

 $\gamma(^{34}\text{Ar})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>
840	4128	3288
1197	3288	2091
1780	3871	2091
2091	2091	0

 $^3\text{He}(^{32}\text{S}, n\gamma)$ 1985A118Level Scheme